Clickbait Without the Click: Linguistic Cues under Temporal Shift Project Proposal for NLP, Winter 2026/27

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Abstract

We investigate whether inexpensive, prespecified linguistic cues improve English clickbait detection beyond fixed sentence embeddings. Three systems are compared: word/character TF–IDF with logistic regression, frozen MiniLM embeddings with logistic regression, and the same embeddings augmented with twelve linguistic features. An audit of the labelled Webis-Clickbait-17 release found 19,538 examples and valid timestamps, but no publisher identifiers. We therefore adopt the topic's permitted time-held-out fallback, with random-split and cross-corpus diagnostics. The deliverables are reproducible splits and models, a controlled feature ablation, and uncertainty, calibration, error, and cost analyses. A negative result is informative if the comparison is properly controlled.

1 Scientific goal and motivation

The input is a social-media teaser (`postText`); the outcome is the published binary `truthClass`, representing perceived clickbait, not factual inaccuracy or observed clicks. We ask: *Does a fixed set of twelve linguistic cues increase out-of-time macro-F_1 relative to exactly the same frozen `all-MiniLM-L6-v2` vectors and logistic-regression classifier?* We also examine random-versus-temporal evaluation and the errors affected by the cues. The hypothesis lacks support if gains disappear after duplicate-aware splitting or are confined to random tests. This is a controlled evaluation rather than a claim to invent feature fusion.

Early studies find informative lexical and stylistic patterns, heavily relying on informality markers or classical feature engineering (Potthast et al., 2016; Biyani et al., 2016). More recently, the field has shifted towards deep learning and large language models; modern representations like Sentence-BERT (Reimers and Gurevych, 2019) and complex hybrid approaches using XGBoost (Michaluk et al., 2026) offer complementary signals but drastically increase computational overhead. However, reported results use different inputs, labels and splits, making scores not directly rankable. Source-separated RoCliCo evaluation (Broscoţeanu and Ionescu, 2023), the Webis annotation study (Potthast et al., 2018) and the assigned seed (Michaluk et al., 2026) motivate our focus on evaluation protocol and isolating the cue contribution. Table 1 records 15 relevant works and the specific published conditions for their outcomes.

2 Data and readiness

Main corpus — READY. We inspected `clickbait17-train-170630.zip` from the official Webis-Clickbait-17 Zenodo release (CC BY 4.0). Its `instances.jsonl` and `truth.jsonl` have matching unique IDs but different row order, so labels must be joined by ID. The 19,538 posts comprise 4,761 clickbait (24.37%) and 14,777 non-clickbait examples. All have timestamps (December 2016–April 2017), whereas publisher IDs and original target URLs are absent. The 38,517-post figure describes the broader corpus, not this ready-to-use labelled archive. We will preserve official binary labels and the five annotator judgments.

The preliminary audit found 54 empty teasers and 411 additional occurrences of normalized nonempty teasers. Removing empties and retaining the earliest normalized occurrence leaves 19,073 examples (4,529 positive), *before* cross-partition article-overlap checks. We will freeze IDs after deduplication and handle repeated underlying article content to prevent cross-split leakage. This source deficiency triggers the explicitly per-

Table 1: Prior evidence. Values are the authors' reported metrics on the specified data and split; different tasks, inputs and averaging conventions prevent direct ranking. CV = cross-validation; dev = development.

Work	Approach and input	Dataset and split	Reported outcome
Potthast et al. 2016	Random forest; tweet features	Webis-16, 2,992 tweets; random 2:1	ROC-AUC .79
Biyani et al. 2016	Boosted trees; title, body, URL	Yahoo, 3,000 train / 1,073 test	Weighted F_1 .749; positive F_1 .603
Anand et al. 2017	Word/character BiLSTM; headlines	15,000 balanced items; 10-fold CV	F_1 .9819
Wei and Wan 2017	SVM/rules and co-training; headlines/articles	Chinese ambiguity (2,924) and misleadingness (2,608); random 3:1	F_1 .753/.724
Potthast et al. 2018	Five-person crowdsourced annotation	38,517 posts, 27 publishers; no model split	Fleiss κ .21 (four levels), .36 (binary)
Clickbait Challenge 2017	Self-attentive BiGRU; strength prediction	Webis-17, 19,538 train / 18,979 test	MSE .033; reported F_1 .683
Indurthi et al. 2020	RoBERTa + boosted regression; teaser only	Same Webis challenge train/test	MSE .0242; reported F_1 .7408
Yi et al. 2022	Contrastive variational text–label model; teaser + body	Webis-derived, 17,588 / 2,000 / 17,554	Macro-F_1 .694
Broscoţeanu and Ionescu 2023	Contrastive Ro-BERT; headline + body	RoCliCo, 6,806 train / 1,507 publisher-disjoint test	Macro-F_1 .9199
Yu et al. 2024	Causal multimodal representation; text, image, profile	Clickbait17; exact split and averaging unspecified	Reported F_1 .9352
Anđelić et al. 2025	Fine-tuned BERTić; Croatian headlines	CLIC, 2,907; 80/10/10	Reported F_1 .776
Michaluk et al. 2026	Embeddings + cues + XGBoost; teaser	40,000 balanced mixed-source items; random 80/10/10	Reported F_1 .864 to .909; width also changes
Reimers and Gurevych 2019	Sentence-BERT + linear classifier	Seven SentEval tasks; ten-fold CV	Mean accuracy 87.41%
Wang et al. 2020	Compact distilled MiniLM	SQuAD2/SST-2 development sets	F_1 72.4 / accuracy 91.0%
Guo et al. 2017	Temperature scaling; TreeLSTM	SST-binary; held-out test	ECE 6.63% to 1.84%

mitted *temporal* rather than unseen-publisher protocol; we will not claim publisher generalization.

External corpus — TO COLLECT. Webis-Clickbait-16 contains 2,992 English tweets, including 767 clickbait cases, with three-annotator majority labels. Acquire and verify its permitted reuse terms, overlap and text fields by 20 November (Adam, supported by Igor); otherwise report why independent transfer could not be measured. No new annotation is planned. This is a diagnostic external comparison, not a second training dataset.

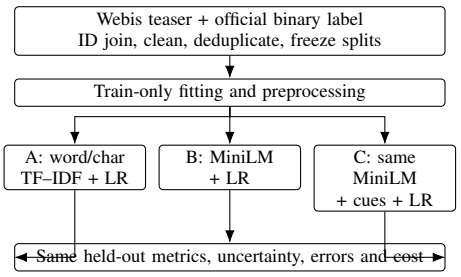


Figure 1: One shared evaluation pipeline. LR = logistic regression; all three branches use the same split IDs and held-out labels.

3 Proposed approach

All systems take identical raw teaser text; no article body, image or publisher metadata enters prediction. We use Python, scikit-learn, sentence-transformers, PyTorch, NumPy and matplotlib; a CPU suffices for linear models, with an optional local GPU for one-time embeddings. The models are: **A:** word (1–2 gram) and character (3–5 gram) TF–IDF + L2 logistic regression; **B:** frozen 384-dimensional `all-MiniLM-L6-v2` embeddings + the same classifier family; **C:** exactly B's cached embeddings concatenated with twelve fixed cues + logistic regression. Features cover length/word structure, punctuation/capitalization and pronoun/forward-reference lexicons. B and C share the encoder, text, split, scaling protocol, validation budget and classifier family, isolating feature value. To account for the severe class imbalance (24.37% positive), all logistic regression models will utilize the `class_weight='balanced'` parameter within scikit-learn. This is the only proposed modelling modification.

4 Evaluation plan

The primary outcome is macro-F_1, with the *paired C minus B* difference as the test of the hypothesis. We additionally report positive-class precision/recall/F_1, average precision (PR ranking), Brier score and reliability diagrams, confusion matrices, computation time and peak memory. Label imbalance makes accuracy insufficient. Calibration and classification must be interpreted

Table 2: Balanced semester workstreams. First names refer to the four named authors.

Workstream	Owner	Support	Main output
Data and protocol	Adam	Igor	Clean joins, leakage-safe time splits, external data
Classical NLP and cues	Hubert	Adam	TF–IDF baseline, cue definitions/ablations
Neural NLP and hybrid	Igor	Mateusz	Frozen MiniLM, embedding and hybrid models
Evaluation and analysis	Mateusz	Hubert	Metrics, uncertainty, error/cost analysis, reproduction

separately (Guo et al., 2017).

Three expanding-origin runs train on December, December–January, and December–February, respectively; the following month is used for validation and the next for testing. The main final comparison uses December–February for training, March for validation and untouched April for testing. We also run three fixed random grouped splits for a protocol diagnostic, and an untuned Webis-16 check if accessible. All preprocessing statistics are fitted on training data. Rather than using a fixed grid, the regularization hyperparameter C will be tuned dynamically on the validation macro-F_1 using randomized search to ensure fair regularization for both sparse (TF-IDF) and dense (MiniLM) representations. We will report individual temporal runs and their spread, a paired bootstrap interval for C–B on April, simple feature-group ablations, and at least 20 categorized misclassifications, including model disagreements and annotator-agreement strata. Hardware, seeds, package versions and commands will be recorded.

5 Work plan and deliverables

Each workstream has one *accountable owner* and one *supporting implementer/reviewer*; the supporting member contributes code, testing or interpretation. All four review approximately four papers each and jointly prepare the report, video, responses to feedback and Q&A. Work division is presented in table 2

Deliverables: (1) documented data loader and immutable split manifests with provenance; (2) reproducible three-system comparison and feature ablation with fixed metrics, uncertainty and compute costs; (3) calibration plots, at least 20 categorized errors, final report and presentation. The minimum viable outcome is a valid A–B comparison even if cues provide no measurable gain. We explicitly exclude spoiler generation, article-body verification, multilingual modelling, browser extensions and live scraping.

Table 3: Weekly plan: owner(s) are accountable for the concrete completion target. Milestone dates follow the course schedule.

Period	Completion target	Owner(s)
30 Oct–4 Nov	Verify sources and data; Phase-1 proposal (4 Nov)	All; Igor edits
5–11 Nov	Loader, splits, TF–IDF; MiniLM cache	Adam, Hubert, Igor
12–18 Nov	Hybrid PoC; feedback (13 Nov); 10-min video (18 Nov)	Igor, Mateusz
19–25 Nov	EDA, evaluation, external-data check, Q&A (25 Nov)	Mateusz, Adam
26 Nov–2 Dec	PoC merge request, revised report, rebuttal (2 Dec)	All; Igor edits
3–9 Dec	Apply feedback; freeze remaining tests	Hubert, Mateusz
10–16 Dec	Main repeated evaluation, CI, 20 errors	Igor, Mateusz
17–23 Dec	External check, figures and draft discussion	Adam, Hubert
24–30 Dec	Break / contingency; no dependency on new work	All if needed
31 Dec–6 Jan	Cross-reproduce best experiment; review final report	All; Adam verifies
7–13 Jan	Final PDF, source/MR, video and Q&A (13 Jan)	All

6 Data statement, limitations, declarations

Data statement. Webis-17 is a public, crowdsourced English social-post corpus distributed under CC BY 4.0; Webis-16 is only a proposed supplementary check pending access/reuse verification. We process teaser text and required ID/time/label fields locally, retain corpus attribution and avoid personal profiles or new scraping. Posts may contain names, handles or links; we do not enrich, rank or publish author-level identities. Released examples follow their respective dataset terms.

Limits and risks. The absent source IDs prevent unseen-publisher claims; temporal tests measure later periods in a historical collection, not deployment in 2026. Subjective labels, remaining topic shifts and possible pretrained-text overlap limit interpretation. The main contingency is to preserve a reproducible temporal rather than invent source groups; unavailable Webis-16 is reported rather than silently replaced.

LLM-use declaration. AI assistants (ChatGPT by OpenAI and Perplexity) were used for literature discovery, initial synthesis of prior work, and language verification/copyediting of the final document draft.

Table 4: Phase-1 contribution record (completed work and actual hours); distinct from the planned semester roles.

Student	Completed Phase-1 work and hours
Igor Kołodziej	Proposal drafting (approach, evaluation, work deliverables); 4 hours
Hubert Kowalski	Proposal drafting (approach, evaluation, work deliverables); 4 hours
Adam Kaniasty	Literature review, source metadata audit and data scoping; 4 hours
Mateusz Iwaniuk	Literature review, source metadata audit and data scoping; 4 hours

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